

Build Event-Centric Soccer Player Identification Pipeline

(Tracking + OCR + Event Clips + Human Review Tool)

OVERVIEW

We are building a lightweight computer vision pipeline to assist human annotators in identifying players involved in discrete on-ball events in full-length 11v11 youth soccer match footage.

This is NOT a full soccer analytics system.

We do NOT need:

- Field geometry or pitch projection
- Ball tracking
- Speed or acceleration metrics
- Tactical or formation analysis

We DO need:

- Player detection
- Multi-object tracking
- Jersey number recognition (best effort)
- Event-level player identification with confidence scoring
- Automatic 7-second event clip extraction
- A simple human-in-the-loop correction interface
- A Dockerized pipeline that we can run locally after delivery

The goal is to correctly identify the primary player involved in each logged event at $\geq 95\%$ confidence (after human review).

1. INPUTS PROVIDED (PER GAME)

We will provide:

- Full-length 11v11 youth soccer match video
- Event Log CSV (approximately 500–1000 events per game) including:
 - Event ID
 - Timestamp

- Team (blue/red)
 - Event type (pass, shot, tackle, etc.)
- Roster CSV per team including:
 - Jersey numbers (starting 11 + bench)
 - Starter vs bench designation
 - Position group (GK / DEF / MID / FWD)

We will provide 5 diverse matches with different cameras, heights, lighting conditions, and jersey contrast quality.

2. CORE REQUIREMENT

We do NOT need all players identified at all times.

We need reliable identification at discrete event timestamps.

For each event, the system must output:

- Primary player jersey number (required)
- Secondary player (if relevant, e.g. assist)
- Confidence score (0–100%)
- Flag if confidence <95% (requires human review)

Primary KPI:

- Percentage of events automatically resolved at $\geq 95\%$ confidence
- After human correction, $\geq 95\%$ confidence for 100% of events

The system may process the full match (or extended windows) to improve identity continuity. For example, if a jersey number is visible 15 seconds before a shot but not at the exact moment, identity should propagate if tracking continuity supports it.

3. REQUIRED DELIVERABLES

3.1 DETECTION AND TRACKING

- Player detection (YOLO or similar)
- Multi-object tracking (ByteTrack, DeepSORT, or similar)
- Team clustering via jersey color

- Per-frame tracking JSON output

3.2 JERSEY NUMBER RECOGNITION

- OCR or number classifier
- Confidence scoring
- Per-track aggregation of jersey hypotheses

3.3 EVENT-LEVEL IDENTITY RESOLVER

- Combine tracking data, OCR output, roster priors, and event timestamps
- Output per-event identity and confidence

3.4 EVENT CLIP EXTRACTION (REQUIRED)

For each event in the Event Log CSV:

- Generate a 7-second MP4
- 2 seconds before event timestamp + 5 seconds after
- One MP4 file per event
- Deterministic naming convention (e.g., event_000123.mp4)
- Output folder containing all clips

3.5 OVERLAY (OPTIONAL BUT PREFERRED)

- Clean overlay showing jersey number and confidence only
- No bounding boxes required in clean mode

3.6 HUMAN-IN-THE-LOOP REVIEW TOOL

- Simple local web interface
- List of events
- Click event -> view corresponding clip
- Display suggested identity and confidence
- Allow override and confirmation
- Save corrections to output file
- Ability to propagate confirmed identities

3.7 SUBSTITUTION HEURISTICS

- Detect appearance of bench players
 - Infer probable substitution window
 - Output incoming jersey, probable outgoing jersey, and time window
-

4. TECHNICAL REQUIREMENTS

- Python-based
- Dockerized or fully reproducible environment
- Must run locally on NVIDIA GPU (e.g., RTX 5060)
- No paid SaaS APIs
- Clear run command (we must be able to process new videos ourselves after delivery)

Example:

```
endvr_cv run --video game.mp4 --events events.csv --roster roster.csv --out output_directory
```

After delivery, we must be able to run this pipeline ourselves without sending videos back to the freelancer.

5. TIMELINE

Expected duration: 1–3 months

Fixed-price preferred.

6. PLEASE INCLUDE IN YOUR PROPOSAL

- Relevant computer vision / tracking projects
- Your approach to jersey number recognition
- How you would compute event-level confidence
- Estimated runtime per full match
- Fixed price quote and milestone breakdown

This project is about building a practical internal tool to accelerate human tagging workflows. Precision at event timestamps matters more than full-match coverage.